POLICY GRADIENT · SOCCER GAME

Policy gradient on the soccer game

We evaluate first-goal and fixed-horizon goal-reset soccer as complementary game formats; neither is designated the primary result. First-goal play is tested on deterministic and random move-order boards, while goal-reset play is tested on the deterministic board. REINFORCE, A2C and PPO use standard training and fictitious play. Results stay separate because a win means a different thing in each format.

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1. Win rates against the best response

Two players face each other. At kickoff one of them has the ball and the other has to defend, and having the ball is an advantage. Each learner is trained for both players, so it is measured in both positions. **Starts with the ball**: our learner is the player who has the ball at kickoff. **Starts without the ball**: our learner is the other player. **Average of both**: the mean of the two, one fair number that does not depend on who starts with the ball.

learner	Deterministic board (A10)				Random move-order board				Goal-reset game, fixed horizon			
	with ball	without ball	average of both	Nash gap	with ball	without ball	average of both	Nash gap	with ball	without ball	average of both	Nash gap
Exact solver	**0%**	**0%**	**0%**	**0.000**	**67%**	**39%**	**53%**	**0.000**	**0%**	**0%**	**0%**	**0.000**
REINFORCE, standard	0%	1%	0%	0.788	13%	0%	7%	0.639	0%	0%	0%	1.139
REINFORCE, fictitious play	2%	1%	2%	0.861	8%	0%	4%	0.745	0%	0%	0%	1.453
A2C, standard	2%	1%	2%	0.763	14%	1%	8%	0.605	0%	0%	0%	1.066
A2C, fictitious play	1%	2%	1%	0.888	12%	1%	7%	0.640	0%	0%	0%	1.469
PPO, standard	0%	0%	0%	0.949	9%	0%	5%	0.750	0%	0%	0%	1.491
PPO, fictitious play	0%	0%	0%	0.864	5%	0%	2%	0.708	0%	0%	0%	1.483
A2C, exact critic, standard	–	–	–	–	–	–	–	–	0%	0%	0%	0.867
A2C, exact critic, fictitious play	–	–	–	–	–	–	–	–	0%	0%	0%	1.435

- **Win rate**: the share of 1,000 games the trained player wins (it scores while the opponent does not; games with no goal or equal goals are ties), when the opponent plays the exact best response to it, the move that minimizes its expected discounted score difference. Mean of 3 seeds.
- The exact solver is the top row, as context rather than a target. The win rate is a check, not an objective: no learner is trained to raise it. Each cell is the mean of 3 seeds.
- **Nash gap (exploitability)** is the sum of both players' gains in expected discounted score if each, in turn, replaces its policy with an exact best response to the other. It is 0 at the exact Nash equilibrium; smaller is closer to equilibrium. This is a payoff gap, not a win rate. The win-rate columns separately show how often the trained player wins against its exact discounted-score best response.
- The random board's kickoff is worth +0.15 to the ball holder even against an exact best response, so the exact solver wins more than half its games when it starts with the ball. On first-goal deterministic play the exact solver ties itself. In goal-reset play, a win means more goals than the opponent over 100 steps. A dash means the learner was not run in that condition.

2. Setup

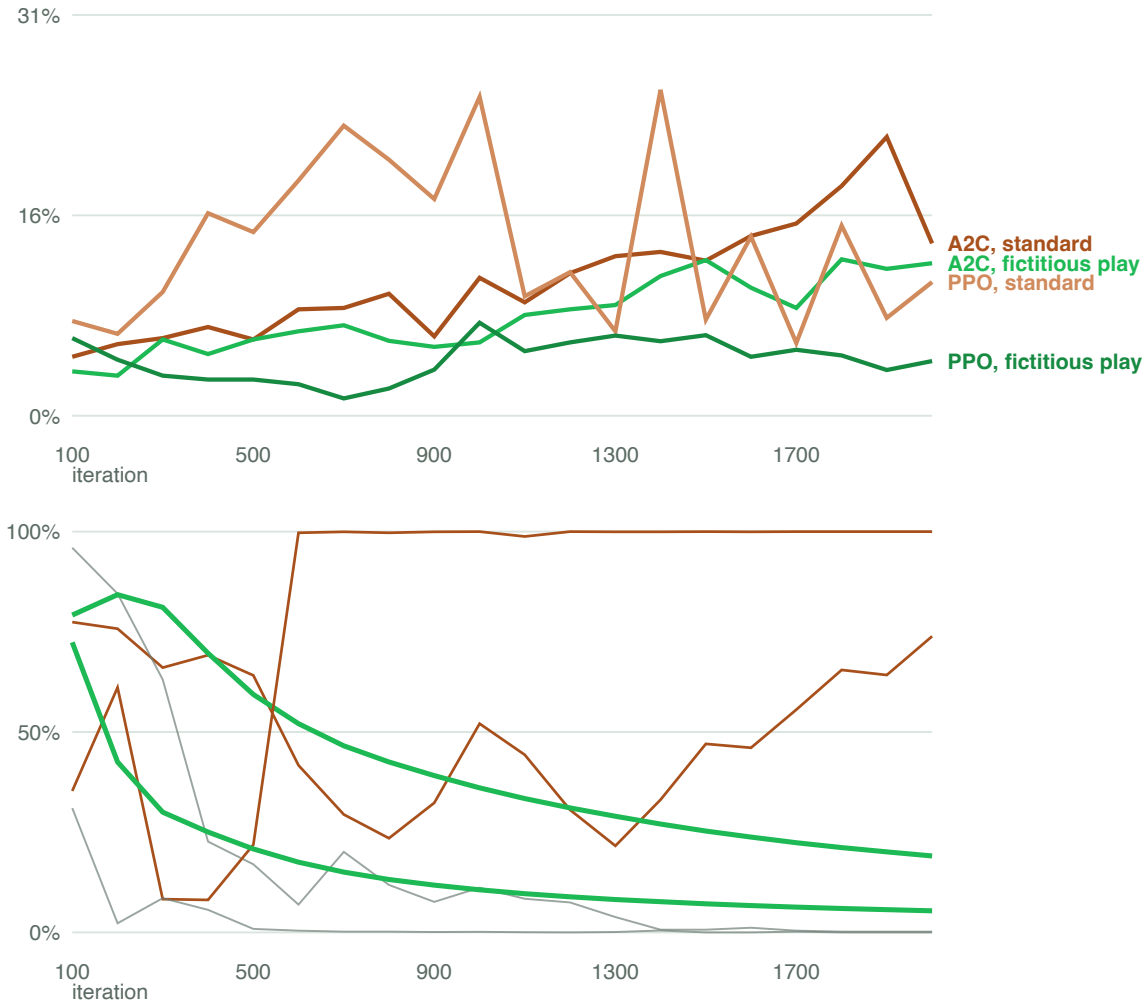
setting	value
Game formats	Both formats are included as complementary conditions, with no primary format: first-goal termination on deterministic A10 and random move-order boards; fixed-horizon goal-reset play on deterministic A10. Each uses a 7×5 board, goal rows 1–3, a 100-step horizon, discount 0.9, and remaining steps as a network input. Results are reported separately: first-goal outcome is who scores first; goal-reset outcome is goal difference over 100 steps.
Algorithms	REINFORCE (discounted returns), A2C (10-step bootstrapped advantage; also with the exact solver's values as a frozen critic), PPO (GAE, clip 0.2, 4 epochs).
Updates	The saved comparisons used one REINFORCE/A2C update per collected batch. The runner now defaults to five shuffled passes over each new batch (--updates-per-batch 5); set it to 1 to reproduce the saved CSVs. PPO keeps its four epochs. Each iteration collects 64 episodes of 100 steps; 2,000 iterations, learning rate 10^{-3}, entropy bonus 0.01, 64×64 networks, separate network per player.
Standard	Both current networks play each other and keep updating.
Fictitious play	Each player trains against a frozen snapshot of the other drawn from all its past snapshots (one every 5 iterations); it reports the average of its own snapshots. Policies, never weights, are averaged.

3. Findings

- **No learner replicates the exact solver.** On the random board the exact solver wins 67% when it starts with the ball, 39% when it starts without the ball and 53% averaged over both starts; the six learners win 5% to 14%, 0% to 1% and 2% to 8%. On the deterministic board the exact solver ties itself and the learners win 2% or less; in the continuing game every learner wins 0%.
- Against a random player the learners win 61% to 96% of games (deterministic board); fictitious play does better than standard training.
- Starting from the 94 states where the exact answer has to mix, the learners win 11% to 15% against the exact Nash policy; the exact solver wins 50%.
- **The two players are not mirror images:** mirror gap 0 for the exact solver, 0.15 to 0.81 for the learners (1 is opposite).
- **Best settings, on 10 seeds:** A2C fictitious play at 10,000 iterations wins 30% when it starts with the ball (95% interval 24% to 36%) against 9% at 2,000; PPO with replayed data wins 28% (13% to 44%) against 10%. Both gains hold up and stay well below the exact solver.

4. Does training converge?

Random board. Every 100 iterations the policy each learner would report is scored against the exact best response (mean of 3 seeds), and the probability it gives to the exact solver's kickoff move (R; the exact solver plays it 100%) is recorded for seed 0. Orange: standard training's current network. Grey: the current network under fictitious play. Green: fictitious play's average.

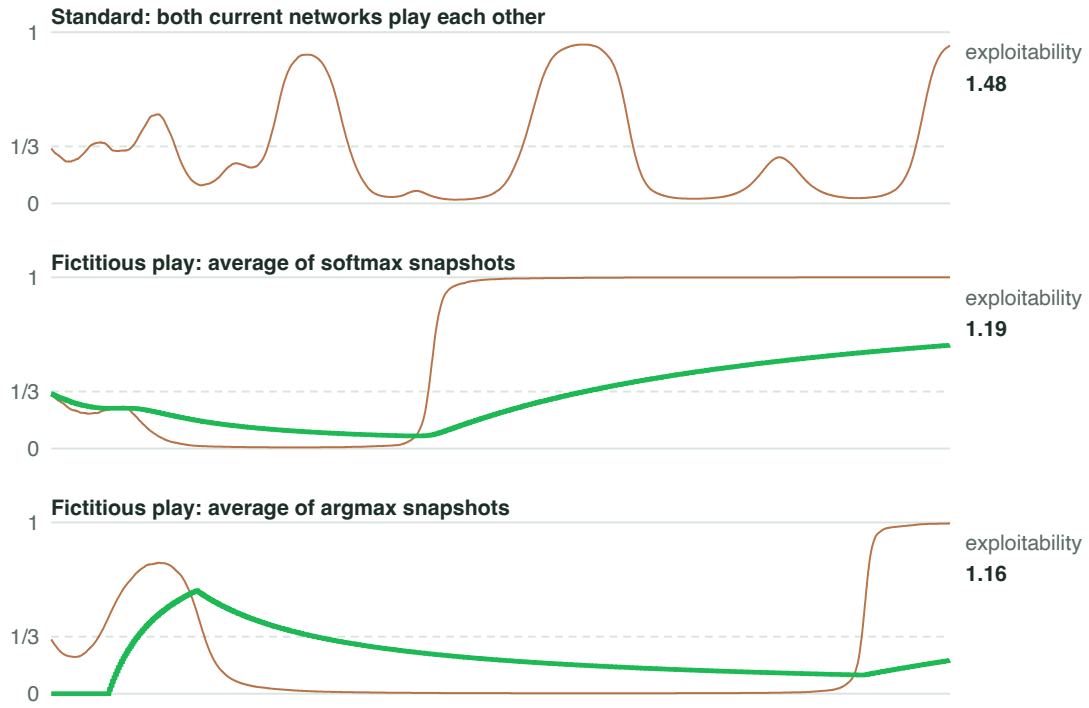


In the second half of training the win rate against the best response stays within A2C standard 9% to 22%; A2C fictitious play 8% to 12%; PPO standard 6% to 25%; PPO fictitious play 4% to 6%; the exact solver wins 67% when it starts with the ball. Standard training keeps swinging; the fictitious-play average is steadier but flat and low.

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5. Rock-paper-scissors with neural networks

Each player is a small softmax network trained by REINFORCE (3,000 iterations, 64 games each, 3 seeds; seed 0 shown). Orange: the current network's share of rock. Green: the average over past snapshots (fictitious play), of softmax policies or of argmax (pure) policies. Exploitability (right) is what a perfect attacker could gain against the two networks' policies, summed over both players: 0 at one third each, 2 at worst. With the soccer settings the standard networks cycle (1.48) and neither average reaches one third (softmax 1.19, argmax 1.16). With a much larger entropy bonus (0.2) and plain gradient steps all three settle: exploitability 0.10 standard, 0.07 softmax average, 0.28 argmax average. That is a stabilization effect of the changed objective, not evidence of convergence to the unregularized equilibrium. Snapshots after 5, 50, 200 iterations of training instead of 5: exploitability 1.23, 1.28, 1.17 (softmax average) and 1.25, 1.29, 1.04 (argmax average).



6. Training choices

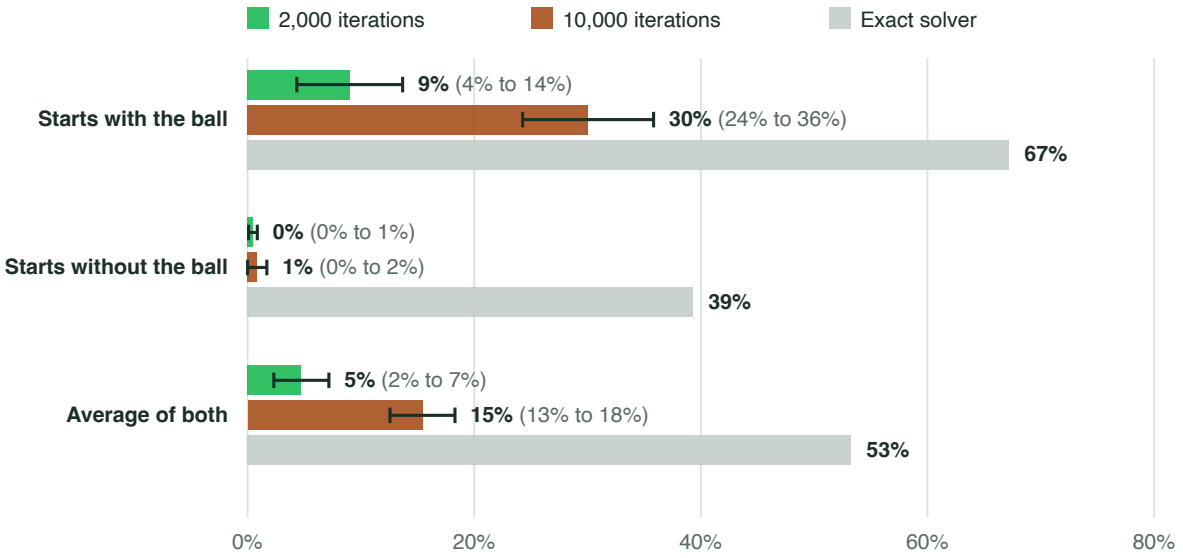
Each comparison changes one thing from the usual setting (2,000 iterations of 64 episodes, a snapshot every 5 iterations, all 100 steps in the loss) on 3 seeds. Changes are in percentage points; single seeds of one setting differ by 10 to 30 points, so differences of that size are noise.

- **One shared network** (two output heads), random board: –3 to +11 points.
- **Last 10 of 100 steps left out**, random board: –8 to +4 points; PPO with standard training stops scoring.
- **Averaging argmax (pure) snapshots** instead of softmax ones: –6 to –4 points on the random board; no gain on the other boards.
- **Entropy bonus 0.2 instead of 0.01** (an ablation; it changes the optimization problem, so it does not show convergence to the unregularized equilibrium), random board: REINFORCE –4, A2C –6, PPO +24 points.
- **Leaving out the last steps: how many?**: win rate, change in points from "all steps": last 5 left out –10 to +11; last 10 left out –8 to +4; last 20 left out –9 to +11; last 40 left out –10 to +13.
- **Leaving out the last steps in the continuing game**: share tied with the exact Nash policy, change in points from "all steps": last 5 left out –59 to –43; last 10 left out –67 to –15; last 20 left out –66 to –44; last 40 left out –40 to –33.
- **How long to train between snapshots**: win rate, change in points from "snapshot every 5 iterations (used elsewhere)": every iteration –6 to –2; every 25 –11 to –3; every 100 –10 to +4.
- **How much data per iteration**: win rate, change in points from "64 episodes per iteration (used elsewhere)": 8 episodes –9 to –3; 256 episodes –4 to +4.
- **Five times as many iterations**: win rate, change in points from "2,000 iterations": 10,000 iterations –9 to +23.
- **Mini-batches and replayed data**: win rate, change in points from "all data, one step": mini-batches of 256 –11 to +6; replay of 4 iterations –2 to +37.
- **The exact solver's values as the critic: random board**: win rate, change in points from "learned critic": exact critic –8 to –5.
- **The exact solver's values as the critic: deterministic board**: share tied with the exact Nash policy, change in points from "learned critic": exact critic –43 to +0.
- **Averaging weights instead of policies**: win rate, change in points from "average of policies": average of weights –7 to –3; average of the output layer's weights –4 to +1.

Ten seeds for the two best-looking settings

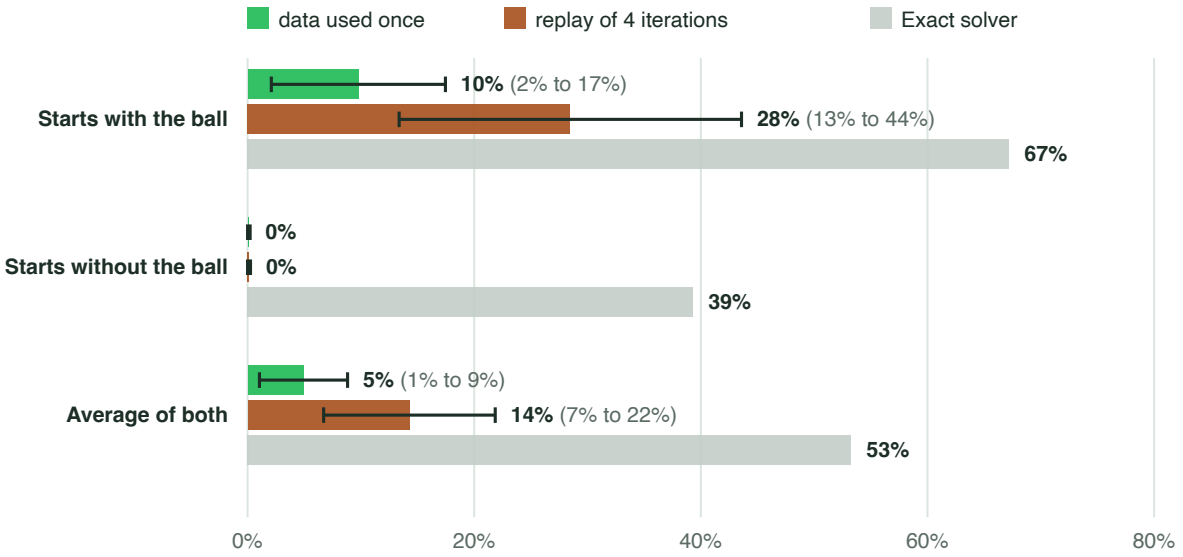
Seeds 0 to 9, random board; 95% intervals (Student's t) in brackets; each bar is the mean of the 10 seeds and the thin line is its 95% interval. The exact solver wins 67% when it starts with the ball, 39% when it starts without the ball and 53% averaged over both starts.

A2C, fictitious play: 2,000 iterations against 10,000 iterations



Starts with the ball 9% (4% to 14%) → 30% (24% to 36%), a change of +21 points (+14 to +28); averaged over both starts 5% → 15%, +11 points (+7 to +14). The interval for the change excludes zero.

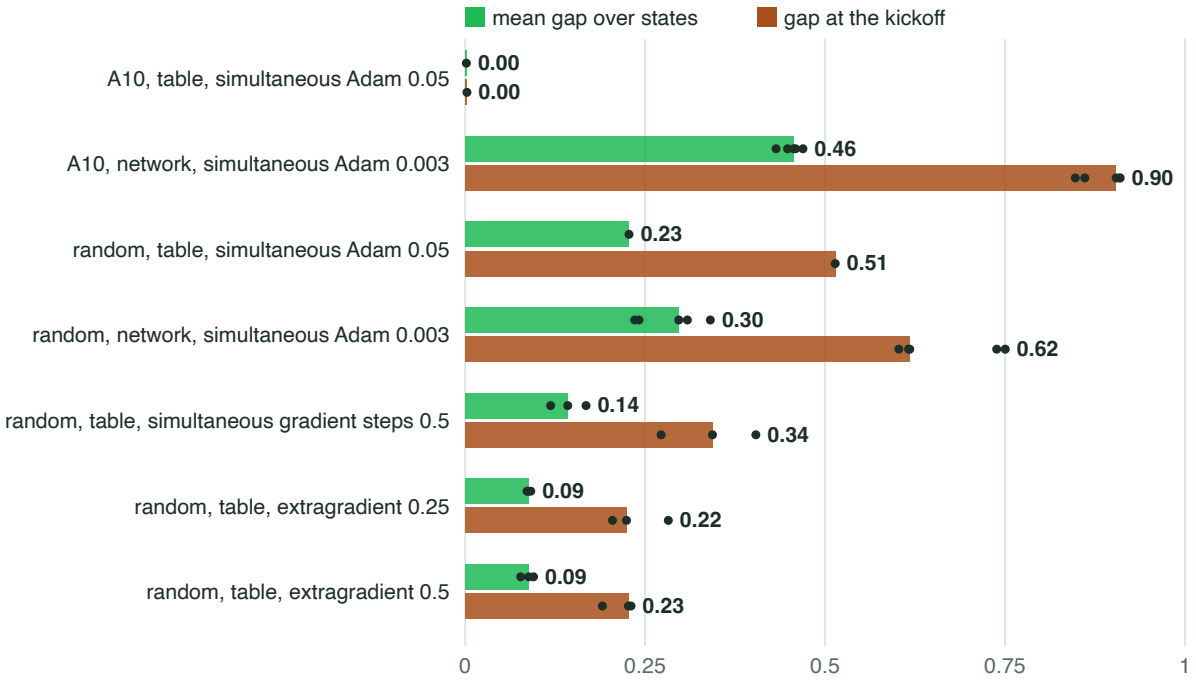
PPO, standard: data used once against replay of 4 iterations



Starts with the ball 10% (2% to 17%) → 28% (13% to 44%), a change of +19 points (+3 to +35); averaged over both starts 5% → 14%, +9 points (+1 to +17). The interval for the change excludes zero.

7. Without sampling noise: exact gradient play

To separate the learning rule from the noise of sampled gradients, both players follow the **exact** gradient of their own value, computed from the transition table on the stationary discounted game (discount 0.9, no step count) and summed over every state, as if a game could start anywhere. The **duality gap** at a state is the first player's best-response value against the second's policy minus the second's best-response value against the first's; it is 0 only at a Markov-perfect (stationary Nash) equilibrium. A **table** has one row of move logits per state; a **network** is one 64×64 network per player, shared across states. Bars are the median over runs (1 table run, 5 network seeds, 3 seeds for the 6,000-step runs) and dots single runs (`scripts/pg_exact_play.py`).



- **Deterministic board (pure equilibrium everywhere): the table reaches the equilibrium and the networks do not.** The table ends with a mean gap of 0.002 and a largest gap of 0.008; every network stalls with a mean gap of 0.43–0.47 and a kickoff gap of 0.85–0.91, the same scale as the sampled learners' exploitability. On this board the network, not the sampling noise, is what stops policy gradient.
- **Random board (the equilibrium mixes at 94 states): no exact gradient method tried here reaches it, even with a table.** The table with Adam stays at a mean gap of 0.23; simultaneous gradient steps leave 0.12–0.17 after 6,000 steps; extragradient, a standard fix for the cycling of simultaneous gradient play near mixed equilibria, roughly halves that to 0.077–0.095 and then stops improving. The networks end at 0.24–0.34.
- So there are two separate obstacles: sharing a network across states stops learning even where the equilibrium is pure, and where it has to mix, simultaneous gradient play does not settle even with a table. Why extragradient stops improving is not tested. The game here is the stationary discounted one, not the finite-horizon games of the sections above.

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8. Related work

Has fictitious play been combined with policy gradient before? Yes, in several forms, and the convergence theory is much older than the neural networks. I checked the titles, authors and venues against Google Scholar (its top result for fictitious play, policy gradient and zero-sum Markov games was Liu, Yang and Wang) and compared each one-line summary with the abstract on the paper's own proceedings, journal or arXiv page, or on a bibliographic mirror where the publisher's page would not load (noted in the entry). JSTOR and the publisher pages for the 1951 papers would not load, so Brown and Robinson are cited from the linked secondary source and were not read. I have not checked the proofs. I did not find a result that proves convergence for deep policy gradient with fictitious play in zero-sum Markov games; the closest results are tabular, regularized or for special game classes.

Older work

- Iterative solution of games by fictitious play (Brown, 1951; Activity Analysis of Production and Allocation, ed. T. C. Koopmans, Wiley, pp. 374-376). Introduced fictitious play as a method for solving two-person zero-sum games (first proposed in 1949); generally regarded as the origin of the method. Not read directly: the publisher and JSTOR pages did not load, so the summary and pages (374-376) come from the linked paper and a university lecture reference list.
- An iterative method of solving a game (Robinson, 1951; Annals of Mathematics 54(2), pp. 296-301). Shows that the iterative procedure suggested by Brown, in which each player in turn plays the best pure strategy against the opponent's accumulated mixed strategy, converges for two-person zero-sum (matrix) games; Shapley (1964) later showed that convergence fails in general games. Not read directly (JSTOR did not load): the summary comes from a Semantic Scholar record and the linked paper.
- Markov games as a framework for multi-agent reinforcement learning (Littman, 1994; ICML, pp. 157-163). Extends reinforcement learning from one adaptive agent to several; studies two agents with directly opposed goals sharing an environment and gives a Q-learning-style algorithm (minimax-Q), shown on a simple game whose best strategy is probabilistic.
- Fictitious play property for games with identical interests (Monderer and Shapley, 1996; Journal of Economic Theory 68(1), pp. 258-265). Defines n-person games with identical interests (best-response equivalent in mixed strategies to a game with identical payoff functions) and proves that every such game has the fictitious-play property. Abstract read on a bibliographic mirror; the publisher's page was blocked.

- Potential games (Monderer and Shapley, 1996; Games and Economic Behavior 14(1)). Defines several notions of potential function for games in strategic form and characterizes the games that have one (in the standard sense that a single function captures every player's incentive to change strategy). Abstract read on a copy of the paper's first page.
- Multiagent learning using a variable learning rate (Bowling and Veloso, 2002; Artificial Intelligence 136(2)). Multiagent learning in stochastic games: a reinforcement-learning technique with a variable learning rate (WoLF, win or learn fast), with a convergence proof in self-play for a restricted class of iterated matrix games and experiments on more general stochastic games.
- On the global convergence of stochastic fictitious play (Hofbauer and Sandholm, 2002; Econometrica 70(6)). Stochastic (perturbed) fictitious play converges globally in four classes of games: those with an interior evolutionarily stable state, zero-sum games, potential games and supermodular games.
- Nash Q-learning for general-sum stochastic games (Hu and Wellman, 2003; Journal of Machine Learning Research 4). Q-learning for general-sum Markov games in which each agent assumes Nash play on the current Q-values; proved to converge under restrictions on the intermediate stage games, which grid-game experiments suggest are not always necessary.
- Generalised weakened fictitious play (Leslie and Collins, 2006; Games and Economic Behavior 56(2)). A broad class of adaptive learning processes extending weakened fictitious play, shown to converge in games with the fictitious-play property; includes an actor-critic process in which a player sees only the reward of the action it chose. Abstract read on the authors' Bristol page.
- Brown's original fictitious play (Berger, 2007; Journal of Economic Theory 135(1)). Brown's 1951 process updates beliefs in a different order from the version used today; with that order, a short convergence proof for nondegenerate ordinal potential games.

Recent work

- Fictitious Self-Play in Extensive-Form Games (Heinrich, Lanctot and Silver, 2015; ICML, PMLR 37). Fictitious play in behavioural strategies of extensive-form games, and a sample-based version (fictitious self-play); approximate Nash equilibria in poker games.
- Deep Reinforcement Learning from Self-Play in Imperfect-Information Games (Heinrich and Silver, 2016; arXiv). Neural fictitious self-play (NFSP) combines fictitious self-play with deep reinforcement learning to approximate Nash equilibria end to end; on Leduc poker it converged near an equilibrium where standard reinforcement learning methods failed to converge, and on Limit Texas Hold'em it came close to leading algorithms that rely on expert knowledge.
- Independent Policy Gradient Methods for Competitive Reinforcement Learning (Daskalakis, Foster and Golowich, 2020; NeurIPS 33). Both players running policy gradient simultaneously reach a min-max equilibrium of a two-player zero-sum stochastic game, provided the learning rates follow a two-timescale rule (which the authors show is necessary).
- Competitive Policy Optimization (Prajapat, Azizzadenesheli, Liniger, Yue and Anandkumar, 2021; UAI 37, PMLR 161, pp. 64-74). Tests game-aware competitive policy gradient (CoPG) and trust-region CoPO against standard gradient-descent-ascent in Markov soccer; CoPG wins more than 74% of reported games against GDA. This is an empirical result for the paper's soccer setup, not a proof of Nash convergence or a training-time guarantee for this project.
- Policy Optimization in Zero-Sum Markov Games: Fictitious Self-Play Provably Attains Nash Equilibria (Liu, Yang and Wang, 2020; OpenReview). A smooth fictitious self-play in which each player takes an entropy-regularized proximal policy optimization (PPO) step against the other; the joint policies converge to a neighbourhood of a Nash equilibrium at a sublinear rate, the first finite-time guarantee for fictitious self-play in zero-sum Markov games (for the regularized method, not the exact equilibrium). Authors and year from the Google Scholar listing and a citation in another paper; abstract read on the Semantic Scholar listing, as the paper's own page did not load.
- Provably Efficient Fictitious Play Policy Optimization for Zero-Sum Markov Games with Structured Transitions (Qiu, Wei, Ye, Wang and Yang, 2021; ICML, PMLR 139). Fictitious-play policy optimization with UCB-style optimism for two-player zero-sum Markov games with structured, unknown transitions (factored independent or single-controller), with tight regret bounds of order the square root of the number of rounds.
- Policy Optimization for Markov Games: Unified Framework and Faster Convergence (Zhang, Liu, Wang, Xiong, Li and Bai, 2022; NeurIPS 35). Each iteration updates the policy at every state with a matrix-game algorithm and then the values; the state-wise average policy converges to an approximate Nash equilibrium of two-player zero-sum Markov games when the matrix-game learners have low weighted regret (rates of roughly T^-5/6, and T^-1 with a modified value update).
- Fictitious play in zero-sum stochastic games (Sayin, Parise and Ozdaglar, 2022; SIAM Journal on Control and Optimization 60(4), pp. 2095-2114). Combines classical fictitious play with Q-learning in two-player zero-sum stochastic games; the strategy beliefs converge to a stationary mixed Nash equilibrium, in both the model-based and model-free cases.
- Fictitious Play in Markov Games with Single Controller (Sayin, Zhang and Ozdaglar, 2022; ACM EC). A new variant of fictitious play for Markov games converges to a Nash equilibrium in n-player identical-interest games where a single player controls the transitions; together with earlier results it covers both zero-sum and single-controller identical-interest Markov games.
- Fast Policy Extragradient Methods for Competitive Games with Entropy Regularization (Cen, Wei and Chi, 2024; Journal of Machine Learning Research 25(4)). Extragradient methods reach the entropy-regularized equilibrium at a linear rate (also in zero-sum Markov games), and tuning the regularization weight approaches an approximate Nash equilibrium of the unregularized game at a sublinear rate.